

# **Session 7: Survival Analysis 2**

Levi Waldron

CUNY SPH Biostatistics 2

# Learning objectives and outline

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## Learning objectives and outline

Review of survival and hazard functions

Cox proportional hazards model

Parametric survival models

- 1 Define proportional hazards
- 2 Perform and interpret Cox proportional hazards regression
- 3 Define time-dependent covariates and their use
- 4 Identify the differences between parametric and semi-parametric survival models
- 5 Identify situations when a parametric survival model might be useful

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Cox  
proportional  
hazards model

Parametric  
survival  
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- ① Review of survival and hazard functions
- ② The Cox proportional hazards model
  - interpretation and inference
  - what are proportional hazards
  - when hazards aren't proportional
- ③ Parametric vs semi-parametric survival models
  - Vittinghoff sections 6.1-6.2, 6.4

# **Review of survival and hazard functions**

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- Study of 6-mercaptopurine (6-MP) maintenance therapy for children in remission from acute lymphoblastic leukemia (ALL)
- 42 patients achieved remission from induction therapy and were then randomized in equal numbers to 6-MP or placebo.
- Survival time studied was from randomization until relapse.

# Leukemia follow-up table

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**Table 3.13** Follow-up table for placebo patients in the leukemia study

| Week of follow-up | No. followed | No. relapsed | No. censored | Conditional prob. of remission | Survival function         |
|-------------------|--------------|--------------|--------------|--------------------------------|---------------------------|
| 1                 | 21           | 2            | 0            | $19/21 = 0.91$                 | 0.91                      |
| 2                 | 19           | 2            | 0            | $17/19 = 0.90$                 | $0.90 \times 0.91 = 0.81$ |
| 3                 | 17           | 1            | 0            | $16/17 = 0.94$                 | $0.94 \times 0.81 = 0.76$ |
| 4                 | 16           | 2            | 0            | $14/16 = 0.88$                 | $0.88 \times 0.76 = 0.67$ |
| 5                 | 14           | 2            | 0            | $12/14 = 0.86$                 | $0.86 \times 0.67 = 0.57$ |
| 6                 | 12           | 0            | 0            | $12/12 = 1.00$                 | $1.00 \times 0.57 = 0.57$ |
| 7                 | 12           | 0            | 0            | $12/12 = 1.00$                 | $1.00 \times 0.57 = 0.57$ |
| 8                 | 12           | 4            | 0            | $8/12 = 0.67$                  | $0.67 \times 0.57 = 0.38$ |
| 9                 | 8            | 0            | 0            | $8/8 = 1.00$                   | $1.00 \times 0.38 = 0.38$ |
| 10                | 8            | 0            | 0            | $8/8 = 1.00$                   | $1.00 \times 0.38 = 0.38$ |

**Figure 1:** leukemia Follow-up Table

This is the **Kaplan-Meier Estimate**  $\hat{S}(t)$  of the Survival function  $S(t)$ .

# Leukemia Kaplan-Meier plot

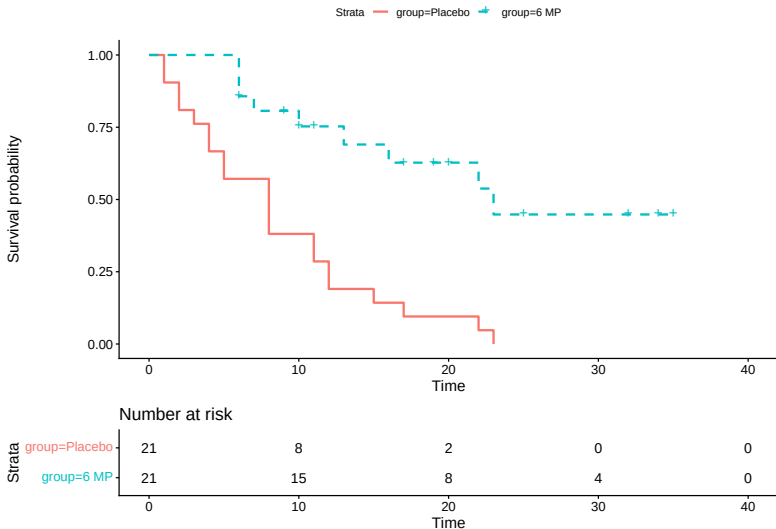
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# The hazard function $h(t)$

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- *Definition:* The *survival function* at time  $t$ , denoted  $S(t)$ , is the probability of being event-free at  $t$ . Equivalently, it is the probability that the survival time is greater than  $t$ .
- *Definition:* The *cumulative event function* at time  $t$ , denoted  $F(t)$ , is the probability that the event has occurred by time  $t$ , or equivalently, the probability that the survival time is less than or equal to  $t$ .  $F(t) = 1 - S(t)$ .
- *Definition:* The *hazard function*  $h(t)$  is the short-term event rate for subjects who have not yet experienced an event.
  - $h(t)$  is the probability of an event in the time interval  $[t, t + s]$  ( $s$  is small), given that the individual has survived up to time  $t$

$$h(t) = \lim_{s \rightarrow 0} \frac{\Pr(t \leq T < t + s | T \geq t)}{s}$$

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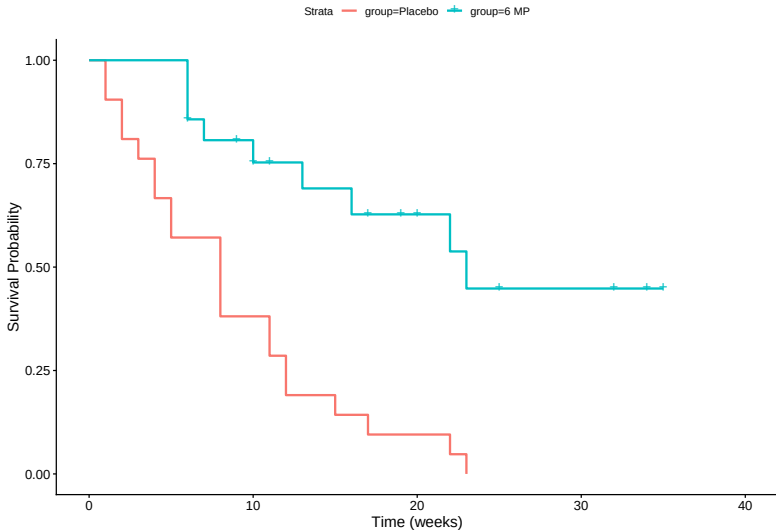
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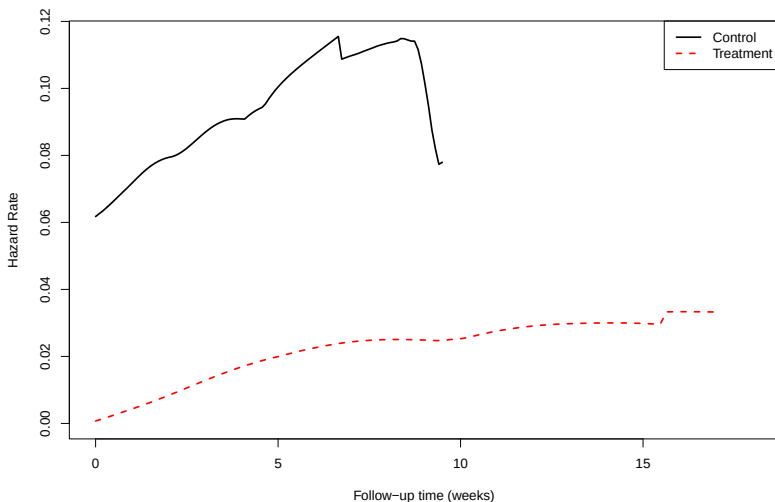
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## Survival function $S(t)$



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## Hazard function $h(t)$



SAS and R Source code

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- If we are comparing the hazards of a control and a treatment group, it could in general be a function of time:
  - $HR(t) = h_T(t)/h_C(t)$
- Interpretation: the risk of event for the treatment group compared to the control group, as a function of time

# The Proportional Hazards Assumption

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- *Definition:* Under the *proportional hazards assumption*, the hazard ratio does not vary with time. That is,  $HR(t) \equiv HR$ .
- In other words,  $HR$  does not vary with time
  - $HR(t)$  is a constant,  $HR$ , at *all times*  $t$
  - this assumption is about the population, of course there will be sampling variation

# A nice proportional hazards dataset

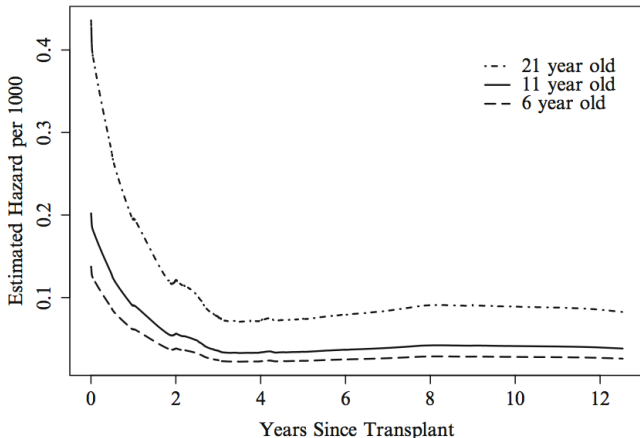
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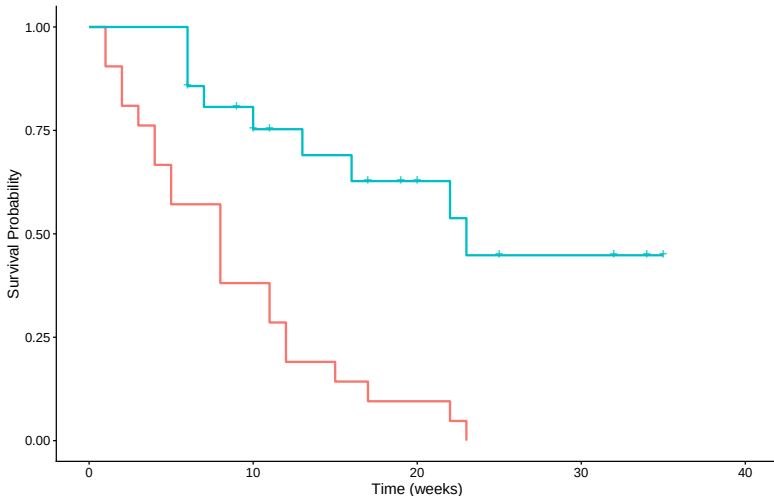


**Fig. 6.3** Hazard functions for 6-, 11-, and 21-year-old transplant recipients

**Figure 2:** Vittinghoff Figure 6.3, p. 210

# The hazard function $h(t)$ (leukemia dataset)

Strata — group=Placebo — group=6 MP



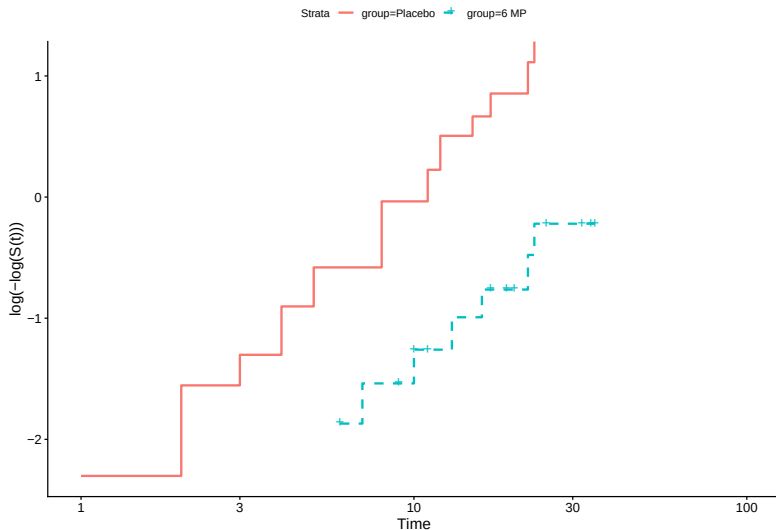
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# Recall previous regression models

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$$E[y_i|x_i] = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi}$$

- $x_p$  are the predictors or independent variables
- $y$  is the outcome, response, or dependent variable
- $E[y|x]$  is the expected value of  $y$  given  $x$
- $\beta_p$  are the regression coefficients

For logistic regression:

$$\text{Logit}(P(x_i)) = \log\left(\frac{P(x_i)}{1 - P(x_i)}\right) = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi}$$

For log-linear regression:

$$\log(E[y_i|x_i]) = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi}$$

# Cox proportional hazards model

# Cox proportional hazards model

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- Cox proportional hazards regression assesses relationship between a right-censored, time-to-event outcome and predictors:
  - categorical variables (e.g., treatment groups)
  - continuous variables

$$\log(HR(x_i)) = \log \frac{h(t|x_i)}{h_0(t)} = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi}$$

- $HR(x_i)$  is the hazard of patient  $i$  relative to baseline
- $h(t|x_i)$  is the time-dependent hazard function  $h(t)$  for patient  $i$
- $h_0(t)$  is the *baseline hazard function*

Multiplicative or additive model?

# Interpretation of coefficients

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- Coefficients  $\beta$  for a categorical / binary predictor:
  - $\beta$  is the *log* of the ratio of hazards for the comparison group relative to reference group ( $\log(HR)$ )
- Coefficients  $\beta$  for a continuous predictor:
  - $\beta$  is the *log* of the ratio of hazards for someone having a one unit higher value of  $x$  (1 year, 1mm Hg, etc)
- If the hazard ratio ( $\exp(\beta)$ ) is close to 1 then the predictor does not affect survival
- If the hazard ratio is less than 1 then the predictor is protective (associated with improved survival)
- If the hazard ratio is greater than 1 then the predictor is associated with increased risk (= decreased survival)

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- Wald Test or Likelihood Ratio Test for coefficients
  - $H_0 : \beta = 0, H_a : \beta \neq 0$
  - equivalent to  $H_0 : HR = 1, H_a : HR \neq 1$
- CIs typically obtained from Wald Test, reported for  $HR$

# Cox PH regression for Leukemia dataset

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```
## Call:
## coxph(formula = Surv(time, cens) ~ group, data = leuk)
##
##   n= 42, number of events= 30
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## group6 MP -1.5721    0.2076   0.4124 -3.812 0.000138 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## group6 MP    0.2076      4.817   0.09251   0.4659
##
## Concordance= 0.69 (se = 0.041 )
## Likelihood ratio test= 16.35  on 1 df,   p=5e-05
## Wald test               = 14.53  on 1 df,   p=1e-04
## Score (logrank) test = 17.25  on 1 df,   p=3e-05
```

# Cox PH is a semi-parametric model

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- Cox proportional hazards model is *semi-parametric*:
  - assumes proportional hazards (PH), but no assumption on  $h_0(t)$
  - robust if PH assumption is not violated
  - time-dependent covariates may resolve apparent violations of the PH assumption.

# Summary: proportional hazards assumption

- Constant hazard *ratio* between groups over time (proportional hazards)
- A linear association between the natural log of the relative hazard and the predictors (log-linearity)
  - A multiplicative relationship between the predictors and the hazard
- Uninformative censoring

# What to do when proportional hazards doesn't hold?

- **Time-dependent covariates**
- **Definition:** A time-dependent covariate is a predictor whose values may vary with time.
- **Basic rule:** You cannot look into the future in your analysis (even though it took place in the past) E.g.:
  - breast cancer chemotherapy patients divided into groups based on how much of the planned dose they received
  - patients divided into groups based on early response to treatment (shrinkage of tumor, lowering of cholesterol, etc)
  - interpolation of the values of a laboratory test linearly between observation times
  - removing subjects who do not finish the treatment plan
  - imputing the date of an adverse event as midway between observation times

Source: Using Time Dependent Covariates and Time Dependent Coefficients in the Cox Model

# Immortal time bias example

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- Immortal time bias is an example of looking into the future.
- E.g. Yee *et al.* reported that new statin users reported a 26% reduction in the risk of diabetes progression with one year or more of treatment relative to never-users (adjusted HR 0.74, 95% CI: 0.56 to 0.97).
  - New users excludes those who had received a lipid lowering drug from three years before to six months after cohort entry
- $HR > 1$  was expected: people whose diabetes progresses are more likely to develop cardiovascular disease, an indication for statins.
- This is a result of an analysis error. Why?
  - Yee *et al.* Statin use in type 2 diabetes mellitus is associated with a delay in starting insulin (<http://onlinelibrary.wiley.com/doi/10.1111/j.1464-5491.2004.01263.x/full>)

# Immortal time bias example (cont'd)

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- What was the analysis error?
  - “new statin user” group was defined based on future initiation: knowledge unknown at the time of entry into the study
  - guaranteed no events for statin users from cohort entry to start of statin use
  - thus all persons in the *treated* group are “immortal” from time 0 until the initiation of statin treatment
  - this period of immortality made treatment look more effective

# Parametric survival models

# What are “parametric” survival models?

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- “Parametric” models estimate additional *parameters* for the baseline hazard, e.g.:
  - **Weibull**: hazard function is a polynomial
  - **exponential**: hazard function is constant over time, survival function is exponential (special case of Weibull):  
e.g. healthy population with randomly occurring events
  - many other options for assumption of distributions
- In most common implementation a log-transform of the time variable is used
  - then can be interpreted as *Accelerated Failure Time* (AFT) models.

# Coefficients in parametric models

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- The interpretation of  $\beta$  coefficients is different:
  - Cox model:  $\log(HR)$
  - AFT models:  $\log(survivaltimeratio)$
  - The sign is *opposite* (i.e. if one is positive the other is negative)

# Why use a parametric survival model?

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- Can be more powerful if assumption is correct
  - may help with small numbers of events
- Extra capabilities:
  - smooth estimation of baseline hazard
  - extrapolation
  - complicated censoring
- Easy to interpret: coefficients are  $\log(\text{survivaltimeratio})$
- Easy to fit: replace `survival::coxph` with `survival::survreg`

# Why not to use a parametric survival model?

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- Depend on correct specification of baseline hazard model
- Even if correctly specified, may not provide much improvement in efficiency
- Still make a proportionality assumption, on survival functions